

SEUPD@CLEF: Team retrix on Source Retrieval for Scientific Web Claims

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Abstract

We present the results of the retrix group for the CheckThat! 2026 - Task 1, in which the goal is to identify the scientific paper implicitly referenced in a social media post. The input consists of short, noisy and usually informal posts that mention scientific findings without providing explicit links or explicit citations.

Our approach is designed to operate in a multilingual setting, as the dataset includes posts in English, German, and French. To handle this, we used multi-lingual language models.

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Keywords

Fact Checking, Information Retrieval, Scientific Document Retrieval, Social media fact verification, CLEF, Neural Re-ranking, Cross Encoder

1. Introduction

The rapid spread of scientific claims through social media creates a difficult fact-checking scenario: posts often report, simplify, or sensationalize research findings while omitting the original source. In many cases, the cited study is not linked explicitly with a URL or a bibliographic reference, but only mentioned indirectly through fragments such as the reported finding, the venue, the institution, or the authors. As a consequence, verifying such claims requires first identifying the scientific publication that is being implicitly referenced.

This problem is the focus of Task 1 of the CLEF 2026 CheckThat! Lab, *Source Retrieval for Scientific Web Claims*. Given a social media post containing a scientific claim and an implicit reference to a paper, the goal is to retrieve the correct publication from a fixed collection of 10.000 candidate papers written in English.

The task is challenging for several reasons. First, the linguistic gap between social media and scientific writing is substantial: posts are short, informal and often contain paraphrases, slogans, hashtags or incomplete statements, whereas scientific papers use more precise and technical language. Second, the reference to the source paper is implicit rather than explicit so the retrieval system must infer the connection from partial semantic evidence rather than from direct citations. Third, relevant clues may appear in different forms, such as mentions of a scientific result, a medical outcome, a research institution, or a publication venue, requiring the system to combine lexical and semantic evidence effectively.

An additional challenge is multilinguality: the official query sets include English, German and French posts while the candidate pool remains shared across languages.

Team Retrix approaches this problem with a three-stage sequential pipeline.

1. Raw tweet-style queries are enriched via pseudo-relevance feedback: BAAI/bge-large-en-v1.5 encodes both queries and corpus documents into a shared embedding space, retrieves the top-5

“Search Engines”, course at the master degree in “Computer Engineering”, Department of Information Engineering, and at the master degree in “Data Science”, Department of Mathematics “Tullio Levi-Civita”, University of Padua, Italy. Academic Year 2025/2026

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semantically similar papers per query and extracts the most frequent terms from those papers as expansion tokens. Not-english queries are processed with BAAI/bge-m3 with the same high-level principles. Not-english queries are translated in English and then processed like the english ones.

2. The expanded queries are then fed into BAAI/bge-m3 operating in hybrid mode, which combines dense inner-product search via FAISS [?] with sparse lexical matching and ColBERT multi-vector scoring to produce an initial ranked list of candidates. The goal here is to have the highest recall possible.
3. Finally, the top candidates are reranked by a neural cross-encoder, in our case nvidia/llama-nemotron-rerank-1b-v2. in our case nvidia/llama-nemotron-rerank-1b-v2 fine-tuned on XXXXXXXX in our case answerdotai/ModernBERT-large fine-tuned on XXXXXXXXXX

Throughout the project we also investigated the effect of fine-tuning the reranker on domain-specific hard negatives extracted from the training set.

Evaluated on the CheckThat! 2026 Task 1 dataset our system ranked XXXXXXXXXXXX on the official leaderboard attaining an average Mean Reciprocal Rank at 5 (MRR@5) of XXXXXXXXXXXX % on the development set (XXXXX place).

The paper is organized as follows: Section 2 discusses the most relevant research directions related to this task; Section 3 describes our approach; Section 4 explains our experimental setup; Section 5 discusses our main findings; finally, Section 6 draws some conclusions and outlooks for future work.

2. Related Work

The task addressed in this paper lies at the intersection of fact-checking and information retrieval. A natural starting point is lexical retrieval, where probabilistic ranking models such as BM25 remain strong baselines thanks to their efficiency, interpretability and robustness in candidate generation [?]. In the setting of scientific claim source retrieval, lexical matching is especially useful when a post mentions recognizable entities or terminology that also appears in the title, abstract, venue or author list of the referenced paper.

Neural retrieval methods have been proposed to mitigate the lexical mismatch between queries and documents by learning dense semantic representations. More broadly, neural information retrieval has shifted attention from exact token overlap to representation learning and learned matching functions [?]. In particular, dense dual-encoder architectures such as Dense Passage Retrieval (DPR) demonstrated that learned query and document embeddings can be highly effective for first-stage retrieval [?]. This line of work is especially relevant to our task because social media posts often paraphrase scientific findings instead of reusing the exact wording found in the original publication.

Between initial retrieval and final ranking, many modern systems adopt multi-stage pipelines that trade off efficiency and effectiveness. Late-interaction models such as ColBERT preserve fine-grained token-level matching while remaining substantially more scalable than fully pairwise neural scoring over the whole collection [?]. Likewise, transformer-based rerankers have become a common second-stage component for refining an initial candidate set with stronger query-document interactions [?]. These approaches motivate retrieval architectures that first generate a manageable set of candidates and then apply more expensive relevance estimation only where it is most useful.

Se fine tuning va bene è da aggiungere parte su hard negatives e ricerca embedding incrociati usate: <https://arxiv.org/html/2505.18366v1> <https://huggingface.co/blog/train-reranker>

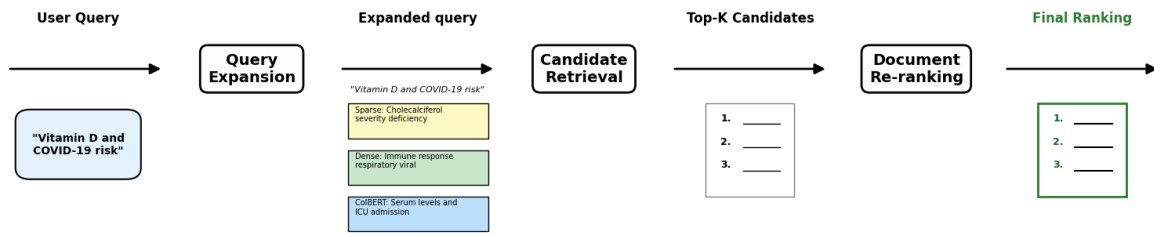


Figure 1: Three-stage pipeline for information retrieval.

Our work is also related to the broader literature on automatic claim verification and, more specifically, to the CheckThat! shared tasks, which have addressed the detection, verification, and analysis of claims in social media across several editions [? ? ?]; however, source retrieval for scientific web claims differs from standard verification tasks in an important way: the system is not asked to classify a claim as true or false, but to recover the original scientific paper implicitly referenced in the post and for this reason, the task is closely connected to evidence retrieval, while also requiring robust matching between informal multilingual posts and formal scientific documents.

3. Methodology

Describe the methodology you have adopted, the architecture of your system, your workflow, etc.

We started by implementing a basic pipeline using BM25 as baseline, using Apache Lucene as the underlying framework. We then experimented with different stop lists, filters and weightings to improve the MRR@5 score, used as final score in the CheckThat! Lab. We then proceeded to experiment with multi-stage pipelines based on neural models, which is the final structure of our project.

We implemented a three-stage pipeline for our information retrieval system, consisting of:

- **Query expansion:** We expanded the original query with additional context to bridge the gap between informal and scientific language.
- **Retrieval:** We retrieved relevant documents from the corpus using the expanded query.
- **Reranking:** We reranked the retrieved documents based on their relevance to the original query.

For each of these stages, we experimented with different models, choosing the ones that performed better in relation to the others.

3.1. Query Expansion

The first step in our pipeline is query expansion: given the original query, we generate extra terms and tokens related to it. This serves two purposes: to increase the number of available tokens for a match, and to bridge the gap between the original informal query and the formal academic language used in the document corpus.

Originally, we experimented with expanding the training dataset queries using **TODO: PRIMO MODELLO USATO (bge m3?)** by performing a generative continuation of the query. We then explored different methods for expansion, including Pseudo-Relevance Feedback (PRF) and similarity with static word embeddings using Word2Vec, testing these on both the original and the preliminary expanded queries.

In the final architecture, we settled on using the **TODO: MODELLO ULTIMO USATO** instruction-tuned model to perform generative augmentation. This model was prompted to generate three distinct representations for the given query, tailored to our multi-stage retrieval needs:

- *sparse*: A list of domain-specific keywords and technical synonyms (e.g., scientific nomenclature) designed to maximize lexical overlap and mitigate the vocabulary mismatch in BM25-based retrieval.
- *dense*: A semantically enriched re-phrasing of the claim that provides a broader context, allowing the bi-encoder in the next step of the architecture to produce more robust embeddings in the high-dimensional vector space.
- *ColBERT*: A structured, formal version of the claim (e.g., "Serum levels and ICU admission") that provides coherent token sequences for fine-grained late-interaction re-ranking.

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This multi-view expansion ensures that the downstream hybrid retriever can leverage both high-precision lexical matches and broad semantic overlaps. By generating these representations simultaneously within a single inference step of **TODO: MODELLO ULTIMO USATO**, we maintain a coherent relationship between the different query 'views', effectively providing a unified but multi-faceted context for the retrieval pipeline.

The resulting expanded queries are then fed into the multi-stage retrieval framework described next.

3.2. Retrieval

3.3. Reranking

3.4. Evaluation

4. Experimental Setup

Describe the experimental setup, i.e.

- used collections
- evaluation measures
- url to git repository and its organization
- hardware used for experiments
- ...

5. Results and Discussion

Provide a summary of the performance on the previous year dataset.

Discuss the results and any relevant issues.

6. Conclusions and Future Work

Provide a summary of what are the main achievements and findings.

Discuss future work, e.g. what you may try next and/or how your approach could be further developed.

7. Group Members Contribution

Fabio Baldan has contributed to this project by:

1. Helping designing and implementing the initial BM25-basic pipeline;
2. Writing and revising some chapters of the documentation based on the actual code and on the concepts learned during the project development;

3. Tracking and reviewing the annotation-part for the runs with various systems;
4. Trying various models for the re-rank part to enhance the MRR@5;
5. Implementing a custom evaluator to focus on metrics that matter the most;
6. Reviewing and improving the query-expansion;
7. Fine-tuning the re-ranker to improve the MRR@5 and assure an advantage if compared with the default one.

Giancarlo Padoan has contributed to this project by:

1. Designing, implementing and fixing the initial basic pipeline;
2. Writing and correcting some chapters of the documentation based on the actual code and on the notions learned while building the project;
3. Designing and implementing an hybrid-retrieval system to enhance recall;
4. Trying various models for the embedding-representation part to enhance the MRR@5;
5. Implementing the first version of query-expansion functionality;
6. Keeping track of the results of the systems that he tried.

Alvise Garberino has contributed to this project by:

1. Helping designing and implementing the initial BM25-basic pipeline;
2. Helping and searching the best models and techniques to implement a re-ranker model;
3. Researching and implementing functionalities for the re-ranker such as data augmentation and data-distillation;
4. Writing some chapters of the documentation providing comprehensive citations and references to the academic material used in the project;
5. Keeping track of the results of the systems that he tried.

Marco Tessari has contributed to this project by:

1. Helping designing and implementing the initial BM25-basic pipeline;
2. Improved the performance of the English Analyzer (text-processing, stemmer, filters and stopwords);
3. Writing part of the docs and helping other members by reviewing their writes;
4. Keeping track of the results of the systems that he tried;
5. Helping and searching the best models and techniques to implement an embedding model;
6. Helping to find the best weights to enhance recall in the hybrid-retrieval system.

Davide Donati has contributed to this project by:

1. Helping designing and implementing the initial BM25 basic pipeline;
2. Implemented a Boolean Query system for the searcher, improving the performance of the retrieval;
3. Writing part of the docs and helping other members by reviewing their writes;
4. Keeping track of the results of the systems that he tried;
5. Improving the multi-lingual query-expansion.

8. Misc [TO BE REMOVED]

INTRODUCTION.TEX E RELATED.TEX SONO DA FINIRE QUANDO AVREMO I RISULTATI COMPLETI (VEROSIMILMENTE QUALCHE GIORNO PRIMA DELLA CONSEGNA) NON CANCELLARE QUESTA FRASE E NON MODIFICARE QUEI FILE (SE NON PER AGGIUNGERE COSE UTILI) SENNÒ MI DIMENTICO DI FARLO E PRENDIAMO 5/20 GRAZIE :) <3



Figure 2: 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (<https://goo.gl/VLCRBB>).

Table 1

Frequency of Special Characters

Non-English or Math	Frequency	Comments
Ø	1 in 1,000	For Swedish names
π	1 in 5	Common in math
\$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

8.1. Tex Files

Put your `TEX` files into the `section` folder as shown in the examples above.

8.2. Figures

Put your figures into the `figure` folder and put the caption under the figure. Example of reference to Figure 2.

8.3. Tables

Put the caption above the table. Example of reference to Table 1.

See the `booktab` packaged documentation for further options.

8.4. Bibliography

Example of citations:

- name: ?]
- parenthesis: [?]

An initial list of references is provided in the files `bibliography.bib` and `proceedings.bib` that you can expand.

See the `natbib` packaged documentation for further options.

8.5. Acronyms

Use the

```
\ac{acronym}
```

command to insert acronyms, eg. *Average Precision (AP)*. The command will expand the acronym the first time it is used.

An initial list of acronyms is provided in the file `acronyms.tex` that you can expand.

See the `acronym` packaged documentation for further options.